KAI WANG

Ph.D. Candidate in Economics & Finance

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Institute of Economic Studies, Charles University



PROFILE

A Ph.D. candidate in Economics and Finance at Charles University, focusing on Financial Econometrics and Asset Pricing. Combining high-frequency econometric modeling with production-grade AI and full-stack software engineering.

WORKING PAPERS & RESEARCH PROJECTS

Gas Fees as Liquidity Risk Signals in Blockchain Token Markets (with Ladislav Krištoufek)

IES Working Paper

2026

- Investigates on-chain network congestion and gas fee dynamics as leading indicators of liquidity risk across blockchain token markets.
- Constructs high-frequency on-chain and market microstructure panels to quantify congestion-induced liquidity premia and spillover channels.

Who Leads on Which Horizon? A Multi-Layer Contagion Map Between DEX and CEX Platform Tokens

Project in progress

2025 – Present

- Constructs a high-frequency connectedness map between DEX and CEX platform tokens across intraday and multi-day horizons.
- Combines TVP–VAR and frequency-domain FEVD spillover measures to distinguish short- and long-horizon leadership and stress episodes.

ACADEMIC POSITIONS

Ph.D. Candidate & Seminar TA (2x “Golden Course” Award)

Institute of Economic Studies, Charles University

Sep 2024 – Present Prague, Czech Republic

- **Econometrics I (JEB109); Econometrics II (JEB110)**
- **Corporate Finance (JEM034); Statistics (JEB034)**

Researcher

Institute of Information Theory and Automation (UTIA), Czech Academy of Sciences

Sep 2023 – Present Prague, Czech Republic

- Contributed to empirical finance projects on crypto-asset and energy markets using advanced econometric and time-series methods.

INDUSTRY EXPERIENCE

AI Software Engineer (Part-time)

Merck Sharp & Dohme (MSD)

Jan 2024 – Present Prague, Czech Republic

Risk Management Analyst (Contractor)

Industrial and Commercial Bank of China (ICBC)

May 2023 – Dec 2023 Prague, Czech Republic

Assistant to Vice President

Winland Group

Mar 2020 – Aug 2021 Beijing, China

EDUCATION

Ph.D. in Economics & Finance

Charles University

Sep 2024 – Present Prague, Czech Republic

Visiting Doctoral Researcher

University of Cologne

Oct 2026 Cologne, Germany

Erasmus Exchange

Johannes Gutenberg University

Sep 2022 – Mar 2023 Mainz, Germany

M.Sc. in Economics

University of Economics, Prague (VŠE)

Sep 2021 – Jan 2024 Prague, Czech Republic

GPA: 1.86

METHODS & TOOLS

- **Backend & Engineering:** Python (`asyncio`, FastAPI, `pytest`), R, Stata, Java OOP, Git.
- **Agentic AI & Eval:** LangGraph, AWS Bedrock AgentCore, A2A streaming, LLM-as-a-Judge, MLFlow.
- **Data & Econometrics:** MySQL, Alembic migrations, high-frequency API pipelines (tick/5-min), TVP–VAR, spectral FEVD, statistical thresholding.

LANGUAGES

- Chinese (native), English (fluent)
- German (A1.1), French (A1).

CERTIFICATES

- GRE General Test: 335/340 (Quantitative 169/170).